ATOMIC-LEVEL MATERIALS SIMULATION

Module Code: MES306TC

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Geometry and Topology of Electronic States

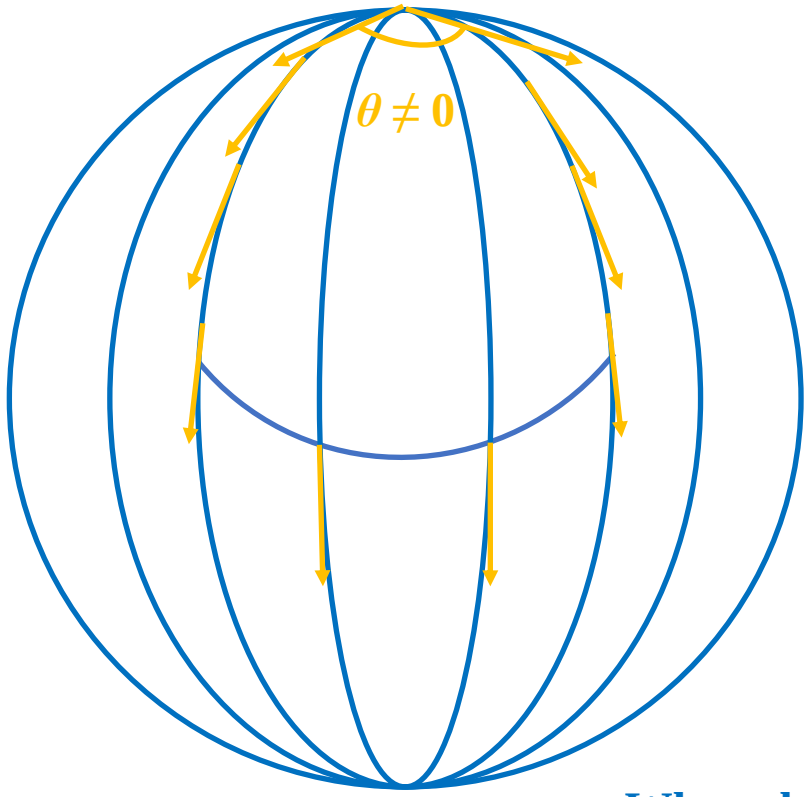
I. Berry Phase and Geometric Effects

II. Topological Insulators and $\mathbb{Z}_2$ Invariants

III. Bulk-Boundary Correspondence

I. Berry Phase and Geometric Effects

Berry Phase — phase acquired by the geometry after the system parameters are varied slowly and brought back to their initial values; it's a memory of the path, not just endpoints



Classic Case



David James Thouless
Nobel Laureate in 2016

wavefunction

Adiabatic evolution: A quantum system is driven slowly (adiabatically) by external parameters, the system remains within its current energy band

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External environmental parameters (*e.g.* the crystal momentum space *k*, an external magnetic field, or an interlayer built-in electric field)

↓

wavefunction evolves continuously with the changing parameters

When the external parameters traverse a closed loop within the parameter space—such as the first Brillouin zone—the system ultimately returns to its initial state. However, in addition to the 'dynamical phase' acquired from time evolution, the wavefunction gains an extra phase determined strictly by the geometric structure of the path. This additional phase is known as the Berry phase

I. Berry Phase and Geometric Effects

change the parameter *k* slowly, the system remains in the instantaneous eigenstate

$|u_n(k)\rangle$ - → $e^{i\gamma} e^{-i\int Edt}|u_n(k)\rangle$

Phase shift is path dependent; $\gamma = \oint A_n(k)\, dk$

Berry connection: $A_n(k) = i\langle u_n(k)|\nabla_k|u_n(k)\rangle$

$A_n(k)$ behaves like a vector potential in the reciprocal space

I. Berry Phase and Geometric Effects

The gauge-invariant field strength: $\Omega_n(k) = \nabla \times A_n(k)$

Reason — **The curl of the gradient of any scalar function is zero**

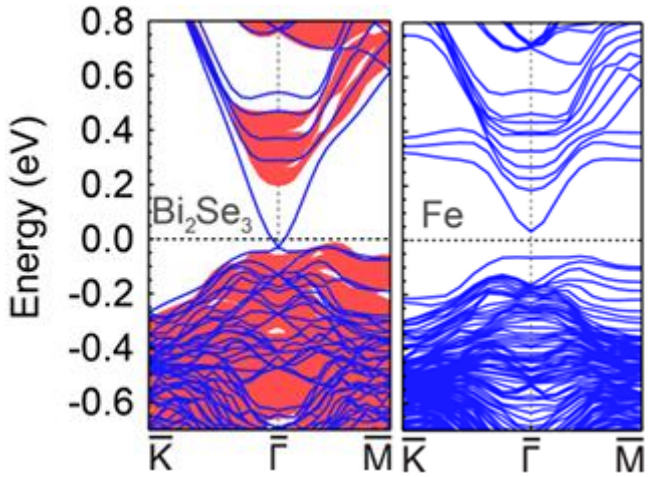
$$\nabla \times \left(A_n(k) - \nabla u_n(k)\right) = \nabla \times A_n(k) - \nabla \times \nabla u_n(k) = \nabla \times A_n(k) = \Omega_n(k)$$

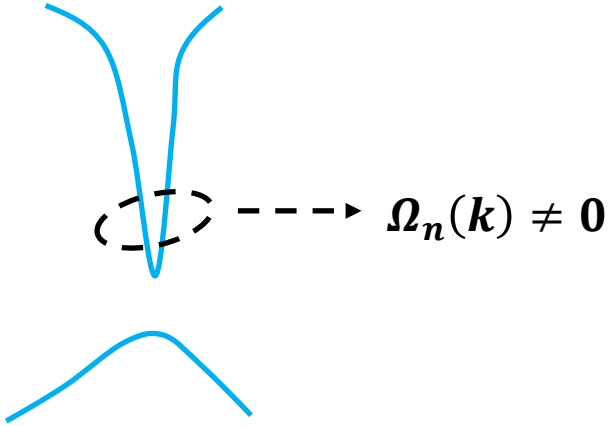
$$\nabla \times \nabla u_n(k) \rightarrow 0$$

$\Omega_n(k)$ is named berry curvature and it acts as a source/sink of flux, characterizing the local geometry of the band

Gap opening with magnetic doping

Band geometry altered





Jeongwoo Kim and Seung – Hoon Jhi, Magnetic phase transition in Fe – doped topological insulator Bi2Se3, Phys. Rev. B, 92,104405 (2015)

I. Berry Phase and Geometric Effects

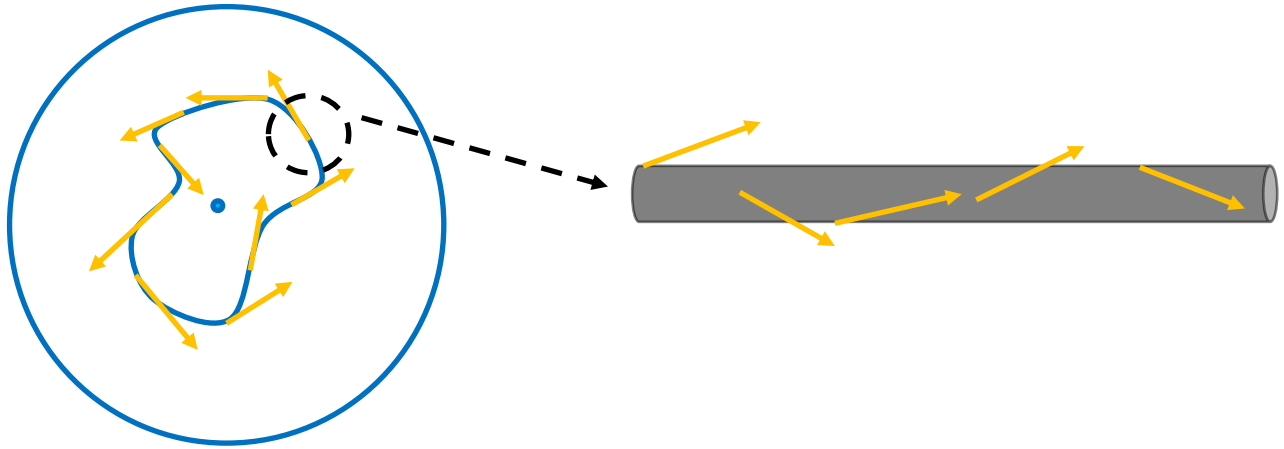
Integrating $\Omega_n(k)$ over the entire Brillouin Zone (BZ) gives the total flux

For a 2D insulator, the integral: $C = \frac{1}{2\pi}\int_{BZ}\Omega_n(k)dk_xdk_y$ is an integer

C — Chern Number and it is a topological invariant

Think of the Berry curvature as magnetic flux coming from a monopole hidden inside the *k* space

Chern Number, *C*, counts how many times the vector field wraps around the sphere

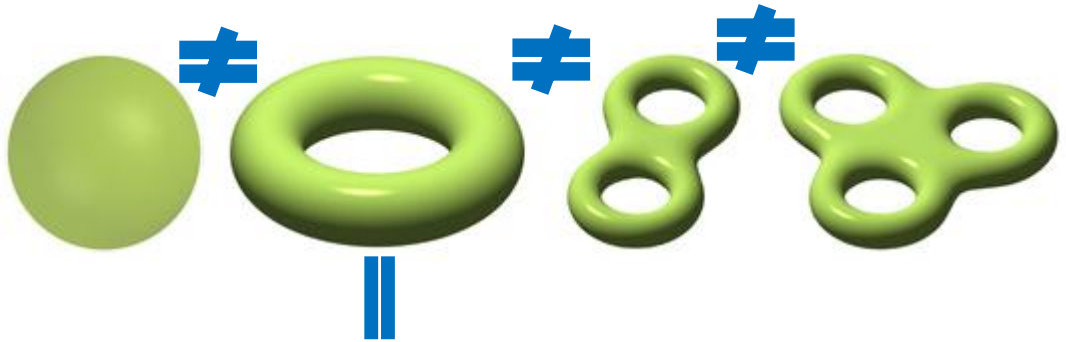


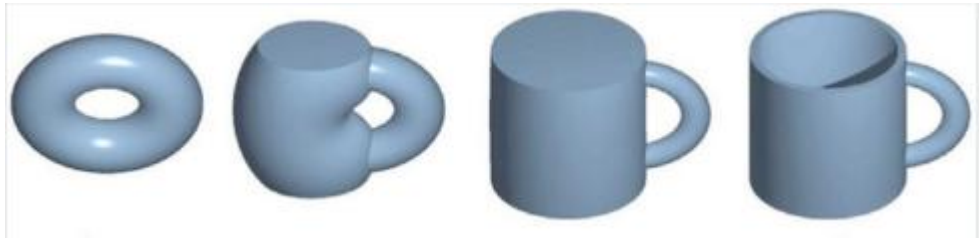
I. Berry Phase and Geometric Effects

The *k* space is a torus (periodic)

Topology classifies these manifolds

If $\Omega_n(k)$ is non-zero, the band structure cannot be 'untangled' without closing the gap





II. Topological Insulators and $\mathbb{Z}_2$ Invariants

What if Time-Reversal Symmetry (TRS) is present? $TH(k)T^{-1} = H(-k)$

↓

This forces the total Chern number to be exactly zero

Electron is fermion and follows $T^2 = -1$ ◂- - - **Kramers Pairs**

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Every state at a Time-Reversal Invariant Momentum (TRIM) point must be at least twofold degenerate

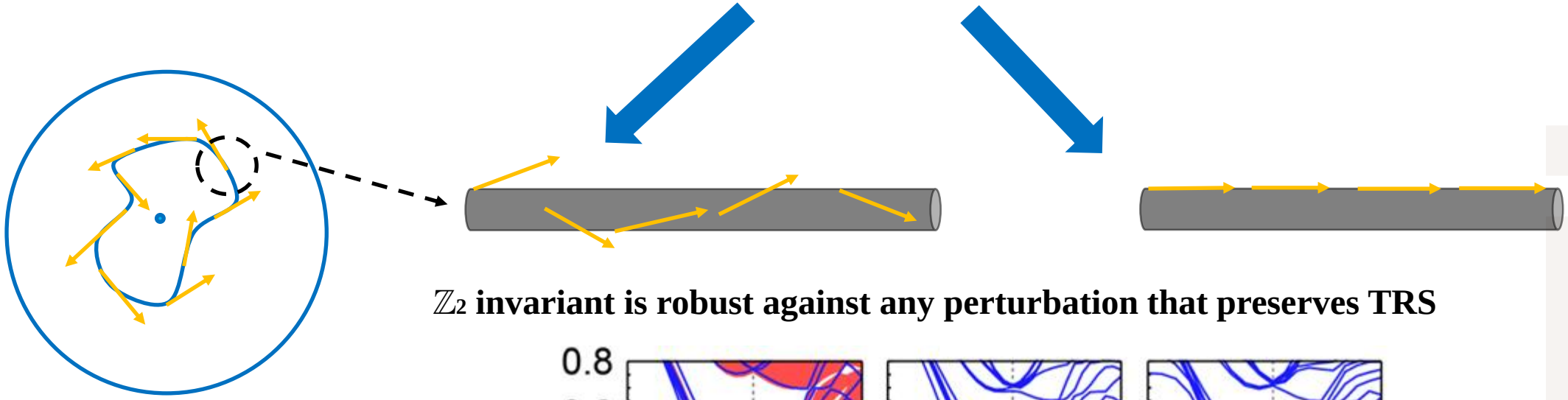
Even if the total Chern number is zero, individual spin sectors can be non-trivial

$\mathbb{Z}_2$ **index** $\nu \in \{0, 1\}$

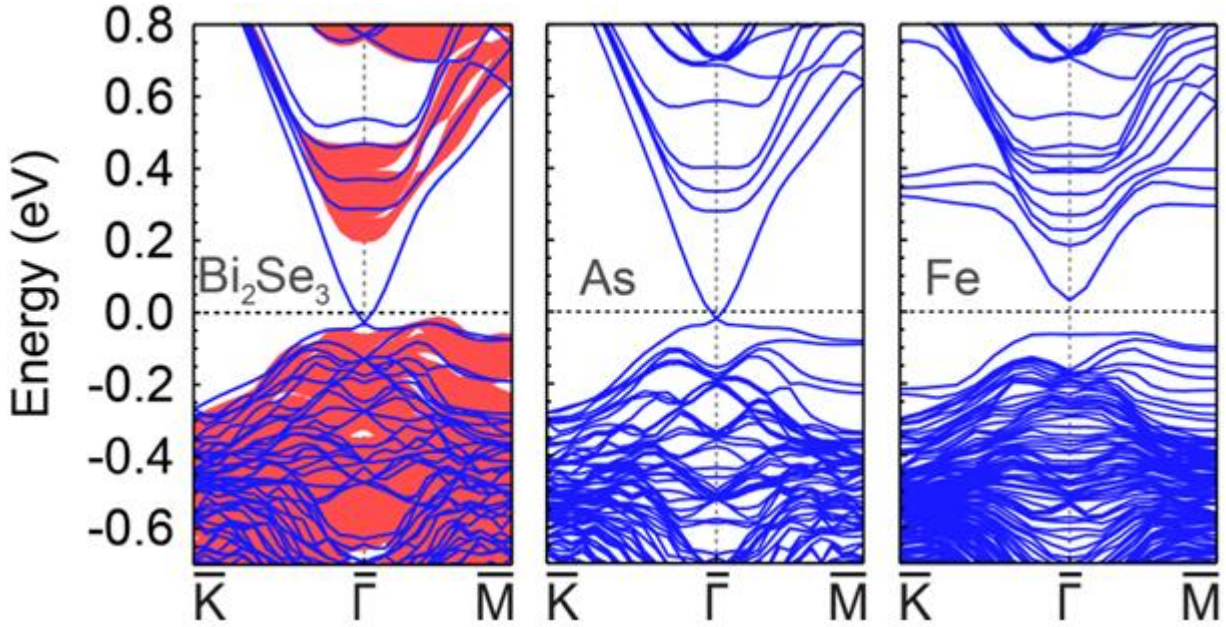
- $\nu = 0$ — **Deformed into a vacuum (Trivial)**
- $\nu = 1$ — **Topologically distinct (Non-trivial)**

II. Topological Insulators and $\mathbb{Z}_2$ Invariants

$\mathbb{Z}_2$ is binary: either it wraps around or it does not



$\mathbb{Z}_2$ invariant is robust against any perturbation that preserves TRS



II. Topological Insulators and $\mathbb{Z}_2$ Invariants

The edge state connects the valence and conduction bands so they are protected

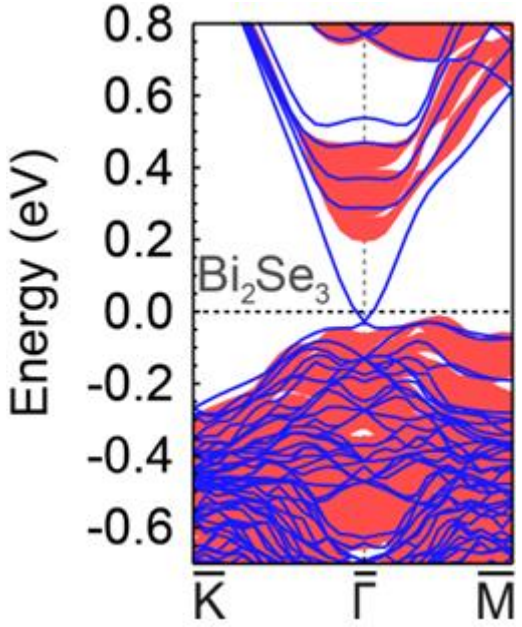
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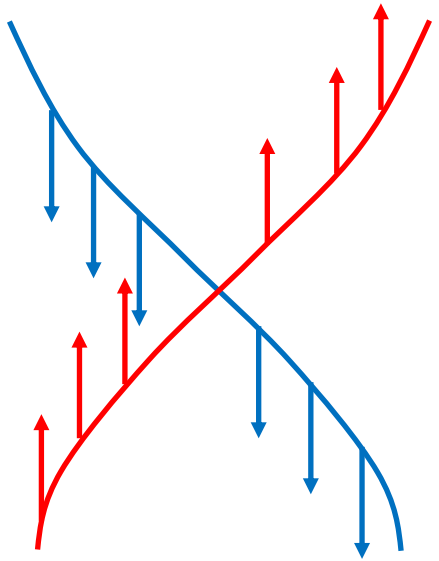
To kill the edge state, you must close the bulk gap (transitioning to triviality)

At the boundary, the two members of a Kramers pair must intersect at TRIM points

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This crossing is robust and protected by TRS





III. Bulk-Boundary Correspondence

The most powerful concept in topological physics

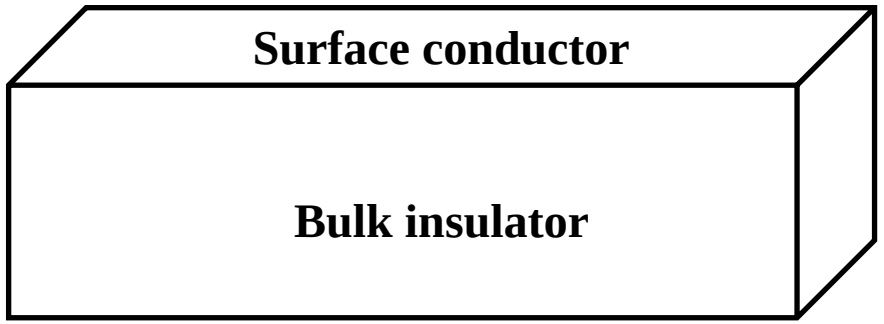
Global bulk properties determine local edge behavior

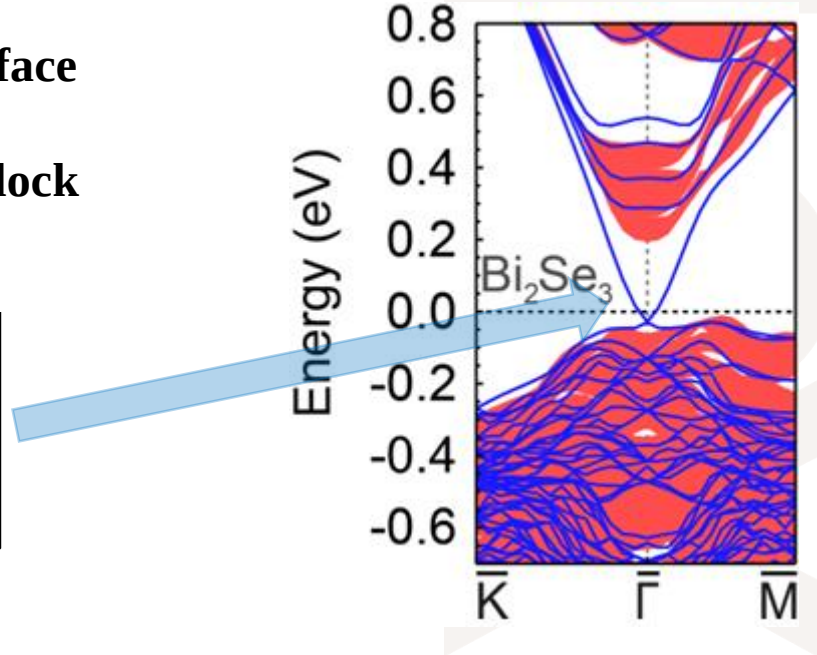
If the Bulk $\mathbb{Z}_2 = 1$, then an Edge state must exist

Think of the interface between a Topological Insulator (TI) and a vacuum (trivial)

The system is forced to close its gap at the interface

The result is a metallic 'skin' on an insulating block



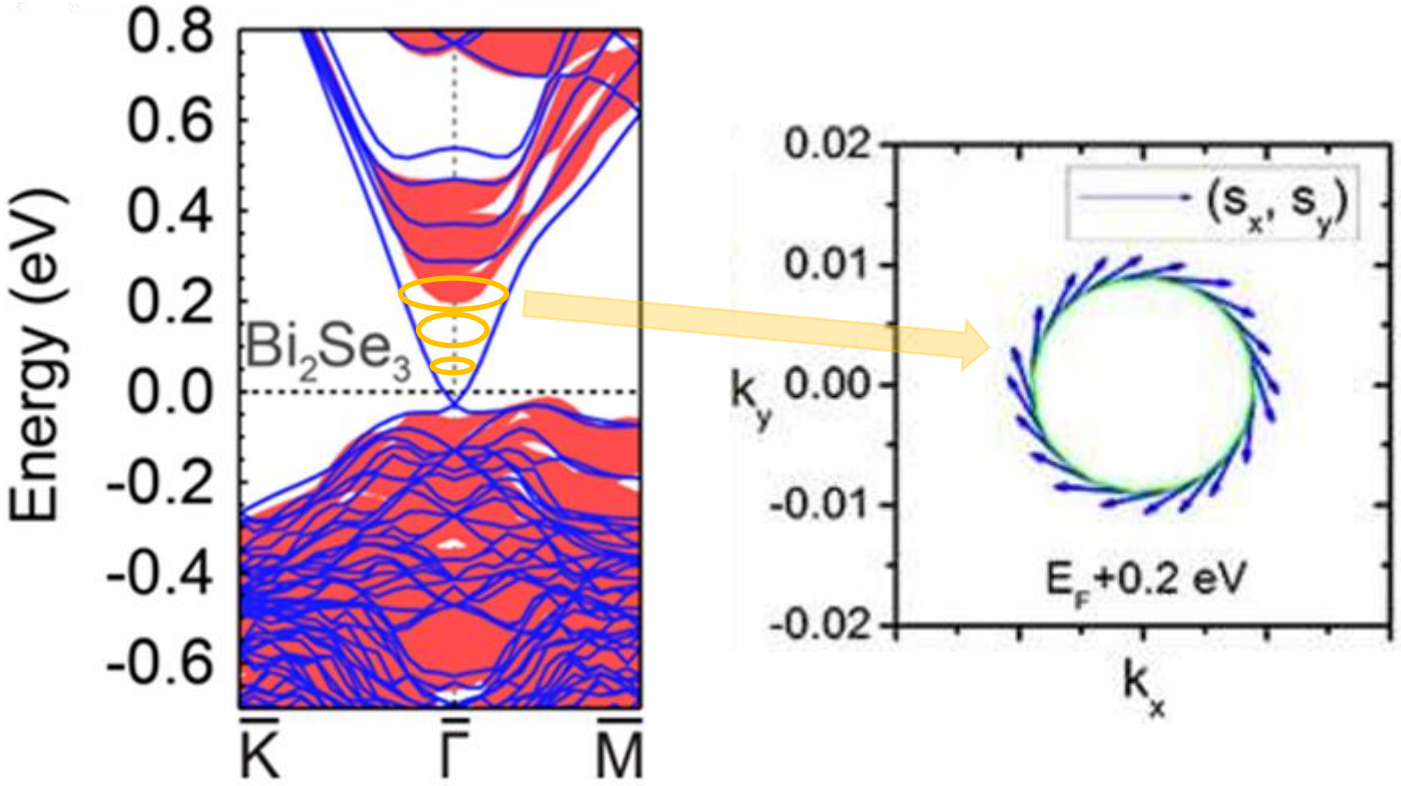


III. Bulk-Boundary Correspondence

The edge states are not just any states; they are spin-polarized

Up-spin moves right; Down-spin moves left

This prevents backscattering from non-magnetic impurities (perfect conduction)

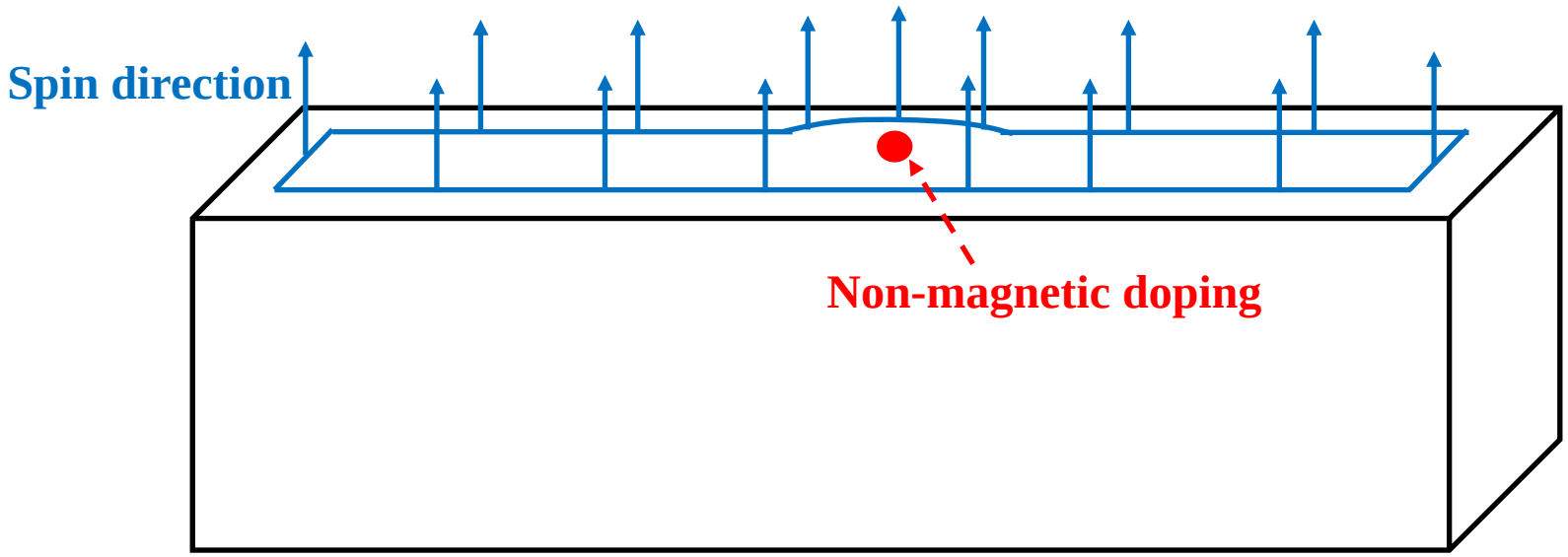


Jia Zhang, Julian P. Velev, Xiaoqian Dang and Evgeny Y. Tsymbal, Band structure and spin texture of Bi_2Se_3 3*d* ferromagnetic metal interface, Phys. Rev. B 94,014435 (2016)

III. Bulk-Boundary Correspondence

Backscattering requires a spin-flip

Without magnetic impurities, spin-flips are forbidden by TRS



III. Bulk-Boundary Correspondence

Trivial (ν = 0): No edge states, gap is everywhere

Non-trivial (ν = 1): Gapless edge states, bulk gap is present

You cannot have a single edge mode without a partner

The partnership is guaranteed by Kramers theorem, ensuring the edge remains gapless



That's all for this session

Thank you for your attention and great participation